Environmental Sensor Analytics

Live DirectQuery over the Kubernetes pipeline · TimescaleDB read replica

Zone

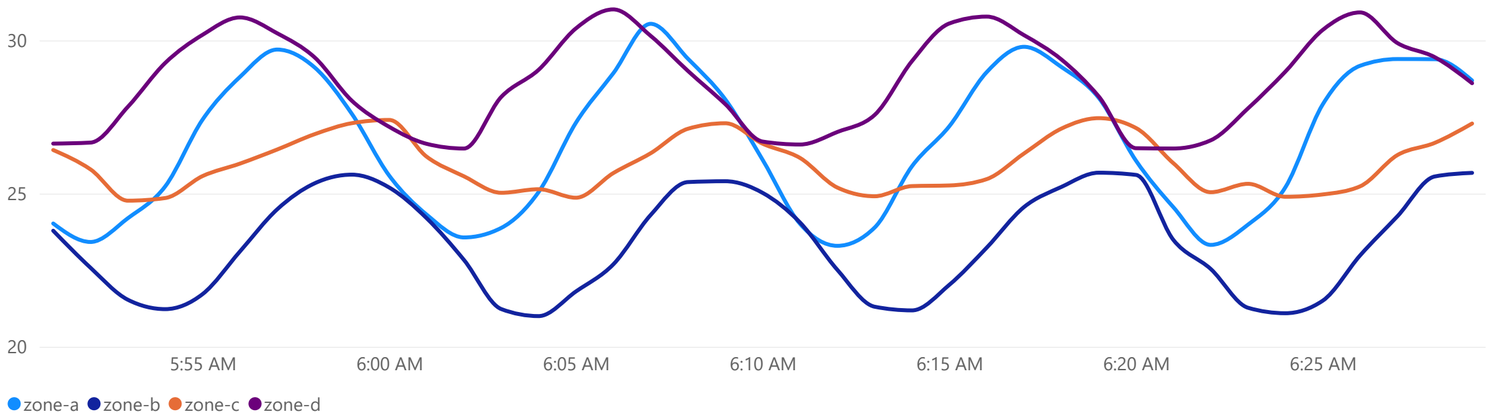
sensor_group

All

120	235.3	82	Moderate	26.2	■ 91 critical	0.002
Sensors Online	Readings per Second	Avg AQI	AQI Category	Avg PM2.5	Alert Status Label	Latency p95 (s)

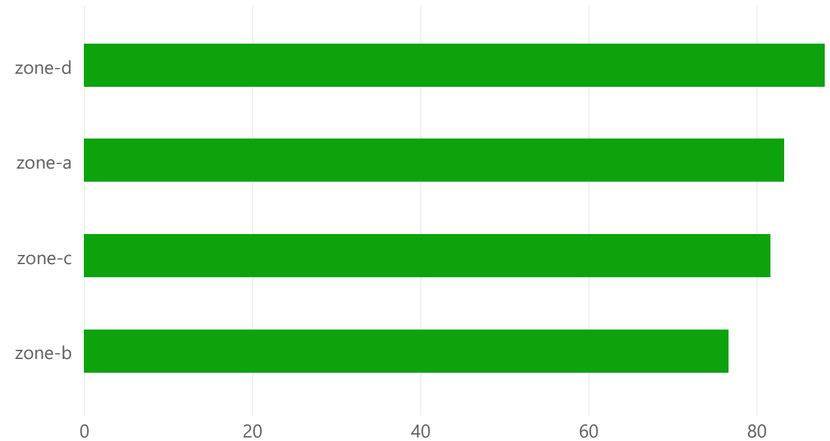
PM2.5 by zone (1-minute average, µg/m³)

Avg PM2.5 by bucket and sensor_group



Average AQI by zone

Avg AQI by sensor_group



Threshold breaches (last 24 h)

bucket	severity	sensor_id	sensor_group	label	value	unit
10/22/2026 6:29:00 AM	critical	node-0103	zone-d	PM2.5	60.0	µg/m³
10/22/2026 6:29:00 AM	warning	node-0002	zone-c	Humidity	82.6	%
10/22/2026 6:29:00 AM	warning	node-0003	zone-d	Humidity	80.9	%
10/22/2026 6:29:00 AM	warning	node-0006	zone-c	AQI	119.6	
10/22/2026 6:29:00 AM	warning	node-0006	zone-c	PM2.5	45.2	µg/m³
10/22/2026 6:29:00 AM	warning	node-0008	zone-a	AQI	146.4	
10/22/2026 6:29:00 AM	warning	node-0008	zone-a	PM2.5	54.1	µg/m³
Total					**155.7**	

Sensor drill-down

Pick a zone or a single sensor; every chart and the table follow

Zone

sensor_group

All

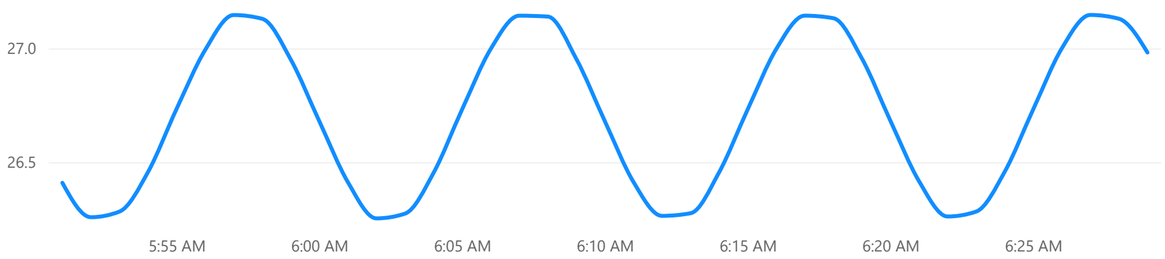
Sensor

sensor_id

All

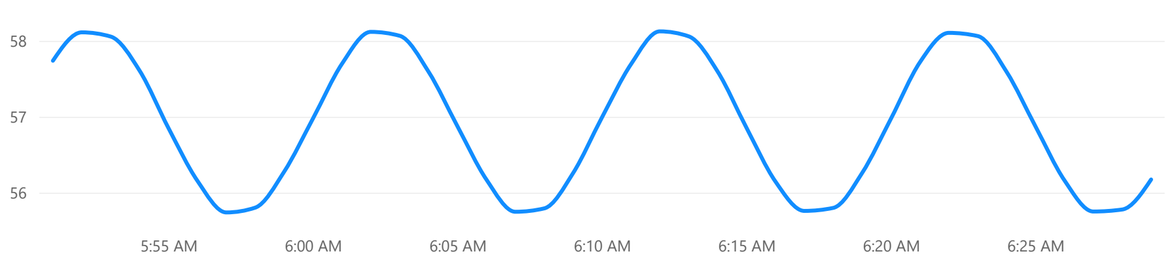
Temperature (°C)

Avg Temperature °C by bucket



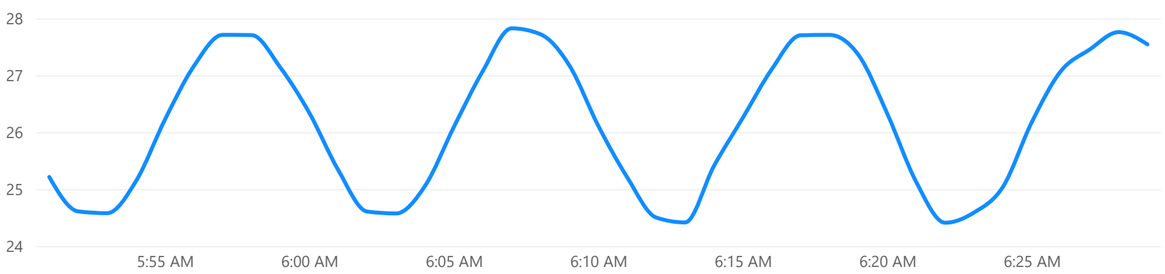
Humidity (%)

Avg Humidity % by bucket



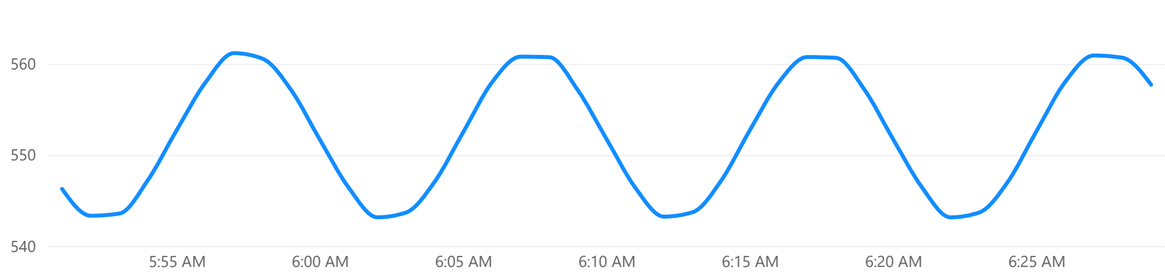
PM2.5 (µg/m³)

Avg PM2.5 by bucket



CO_2 (ppm)

Avg CO2 ppm by bucket



Latest reading per sensor (worst AQI first)

sensor_id	sensor_group	ts	aqi	aqi_category	pm25	pm10	temperature_c	humidity	co2	is_anomaly
node-0029	zone-b	10/22/2026 6:29:44 AM	152	Unhealthy	56.84	88.54	27.86	41.50	681.12	False
node-0075	zone-d	10/22/2026 6:29:43 AM	150	Unhealthy for Sensitive Groups	55.22	95.11	34.30	61.16	552.07	False
node-0101	zone-b	10/22/2026 6:29:44 AM	148	Unhealthy for Sensitive Groups	54.43	101.48	26.70	48.94	547.16	False
node-0008	zone-a	10/22/2026 6:29:44 AM	146	Unhealthy for Sensitive Groups	53.94	77.54	26.07	55.03	573.80	False
node-0105	zone-b	10/22/2026 6:29:44 AM	146	Unhealthy for Sensitive Groups	53.78	97.86	29.93	67.05	531.85	False
node-0034	zone-c	10/22/2026 6:29:44 AM	136	Unhealthy for Sensitive Groups	49.63	70.57	28.92	48.40	565.39	False
node-0067	zone-d	10/22/2026 6:29:43 AM	132	Unhealthy for Sensitive Groups	47.94	90.06	30.93	60.32	530.28	False
node-0032	zone-a	10/22/2026 6:29:44 AM	129	Unhealthy for Sensitive Groups	46.99	71.86	26.52	57.29	606.85	False

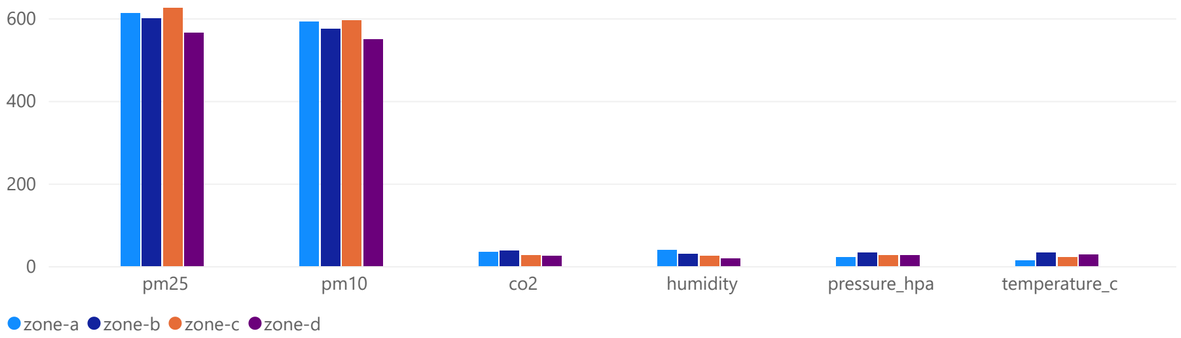
Anomalies and threshold breaches

Anomalies: per-sensor EWMA z-score > 4 · Breaches: 1-minute averages over the thresholds table

2898	0.52%	69	90
Anomalous Readings	Anomaly Rate	Sensors in Breach	Critical Breaches

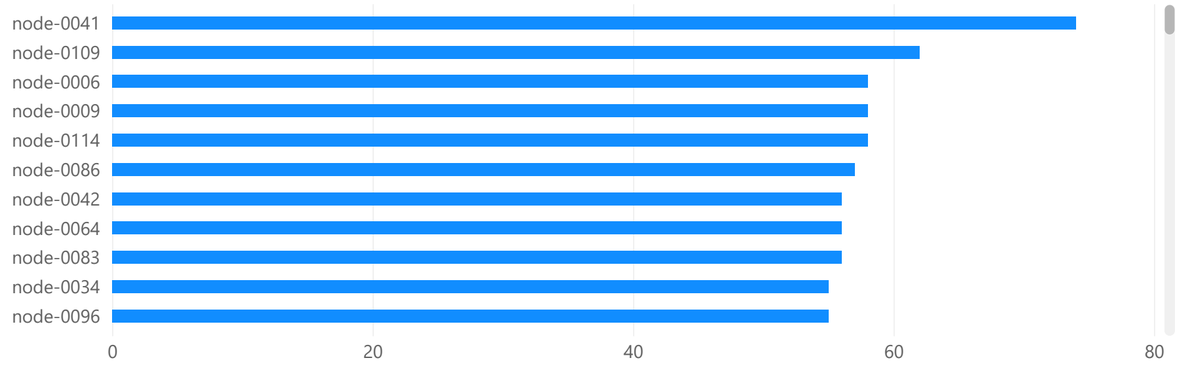
Anomalies by metric and zone (last 24 h)

anomalies by metric and sensor_group



Anomalies per sensor (most first)

anomalies by sensor_id



Anomalous readings

ts	sensor_id	sensor_group	metric	pm25	pm10	aqi
10/22/2026 6:29:46 AM	node-0097	zone-b	pm10	225.5	360.8	300
10/22/2026 6:29:46 AM	node-0097	zone-b	pm25	225.5	360.8	300
10/22/2026 6:29:45 AM	node-0004	zone-a	pm10	190.8	305.3	266
10/22/2026 6:29:45 AM	node-0004	zone-a	pm25	190.8	305.3	266
10/22/2026 6:29:44 AM	node-0043	zone-d	pm10	108.1	173.0	188
10/22/2026 6:29:44 AM	node-0043	zone-d	pm25	108.1	173.0	188
10/22/2026 6:29:43 AM	node-0032	zone-a	pm10	382.8	612.5	500
10/22/2026 6:29:43 AM	node-0032	zone-a	pm25	382.8	612.5	500
10/22/2026 6:29:43 AM	node-0000	zone-a	pm10	297.7	476.3	445
10/22/2026 6:29:43 AM	node-0000	zone-a	pm25	297.7	476.3	445
Total				**663.7**	**1062.0**	**500**

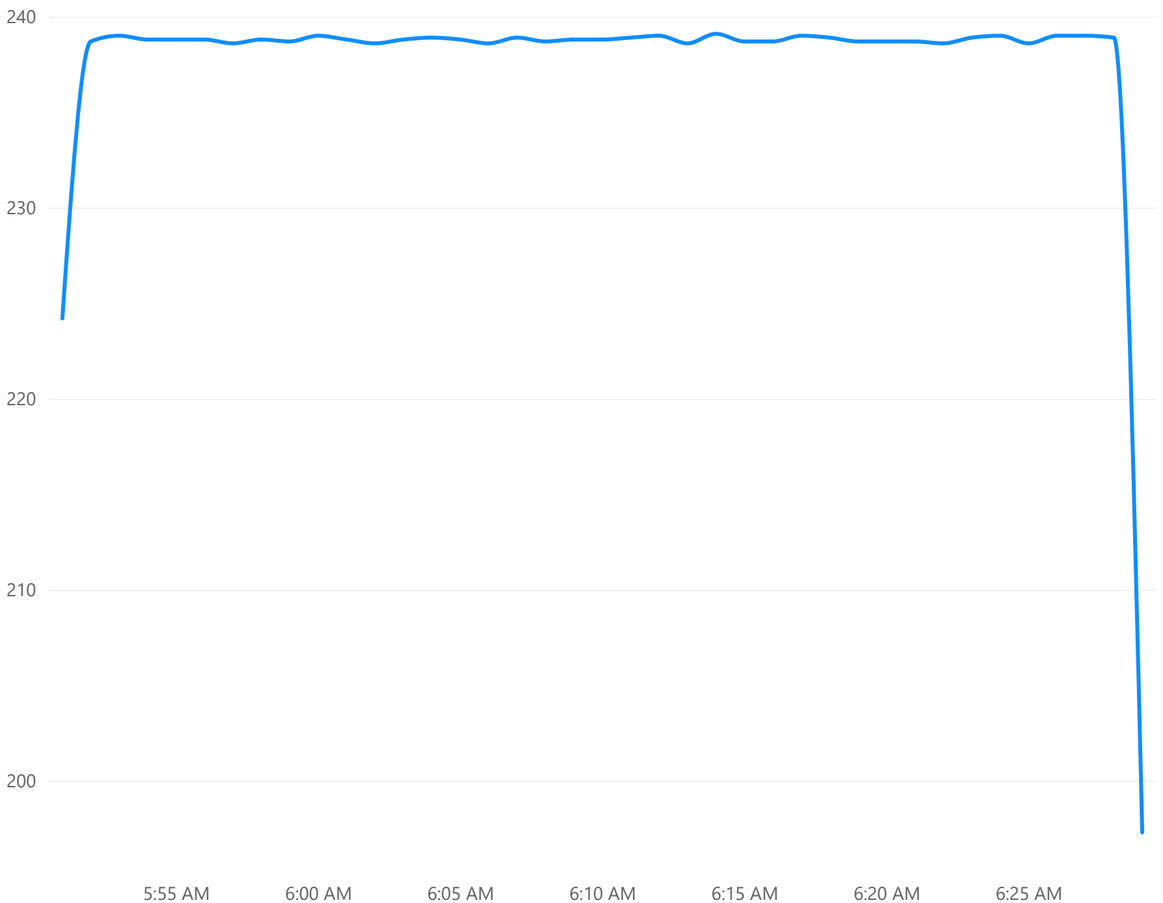
Pipeline health

Throughput and end-to-end latency (sensor timestamp → database commit)

72K	237.3	0.001	0.002	2096	385
Readings (last 5 min)	Pipeline Throughput (rec/s)	Latency p50 (s)	Latency p95 (s)	Gap-filled (last 5 min)	Anomalies (last 5 min)

Records committed per second

records per second by minute



End-to-end latency percentiles (seconds)

p50, p95 and p99 by minute

